



SOLVED QUESTIONS FOR FINAL TERM EXAM [LECT 09 TO 15]

L09&15_Q01: Find the numerical plot of the following initial value problem on [0,2]:

$$y'(x) = y - x^2 + 1, y(0) = 0.5$$

%Find the numerical plot of the following initial value problem on [0,2]:

```
% y'(x)=y-x^2+1 , y(0)=0.5
clc
clear all
close all
f=@(x,y) (y-x.^2+1+x*y)
ode45(f,[0,2],0.5)
xlabel('x-axis')
ylabel('y-axis')
title('Numerical Plot of IVP')
grid
```

L09&15_Q02: Plot the family of the approximate solutions of the following Initial Value Problem (IVPs): $y'(x) = y - x^2 + 1, y(0) = 0.5:0.3:3$

%Plot the family of the approximate solutions of the following Initial Value Problem (IVPs):

```
% y'(x)=y-x^2+1 , y(0)=0.5:0.3:3
clc
clear all
close all
f=@(x,y) (y-x.^2+1)
ode45(f,[0,2],0.5:0.3:3)
xlabel('x-axis')
ylabel('y-axis')
title('Family of the approximate solutions of IVPs')
grid
```

L09&15_Q03: Compute the numerical values of the approximate solution of the following IVP at $x=0:0.2:2$, $y'(x) = y - x^2 + 1, y(0) = 0.5$

%Compute the numerical values of the approximate solution of the following IVP

```
% at x= 0:0.2:2, y'(x)=y-x^2+1 , y(0)=0.5
```

```
clc
clear all
close all
f=@(x,y) (y-x.^2+1)
[x,ya]=ode45(f,0:0.2:2,0.5)
plot(x,ya)
```

L09&15_Q04: Find the exact solution of the following IVPs, then compute its approximate solution at $x=0:0.2:2$ using ode45 command. Finally, makes the table of values at t,ya and y(exact solution).

$$y'(x) = y - x^2 + 1, y(0) = 0.5$$

%Find the exact solution of the following IVPs,

%then compute its approximate solution at $x=0:0.2:2$ using ode45 command.

%Finally, makes the table of values at t,ya and y(exact solution).

```
%y'(x)=y-x^2+1 , y(0)=0.5
clc
clear all
close all
eq='Dy=y-x^2+1';
initialcond='y(0)=0.5';
y=dsolve(eq,initialcond,'x')
f=@(x,y) (y-x.^2+1)
[x,ya]=ode45(f,0:0.2:2,0.5)
[x,ya,2.*x-exp(x)./2+x.^2+1]
```

L09_Q05: Find the exact solution of the following IVPs at some specific domain elements, $x=-0.2,-0.4,-0.6,-0.8,-1.0$ in the interval $[-1,0]$ using ode45 command.

$$y'(x) = y - x^2 + 1, y(0) = 0.5$$

% Find the exact solution of the following IVPs at some specific domain elements,



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```
% x=-0.2,-0.4, -0.6, -0.8, -1.0 in the interval [-1,0] using ode45 command.
% y'(x)=y-x^2+1, y(0)=0.5
clc
clear all
close all
f=@(x,y) (y-x.^2+1)
ode45(f,0:-0.2:-1,0.5)
xlabel('x-axis')
ylabel('y-axis')
title('Graphical view of IVP when initial condition is mentioned at the right end
pont')
grid
```

L09_Q06: Plot the family of the approximate solutions of the following IVP on the interval [-1, 2] using ode45 command. $y'(x) = y - x^2 + 1, y(0) = 1:0.2:3$

```
%Plot the family of the approximate solutions of the following IVP on the interval
[-1, 2]
% using ode45 command. y'(x)=y-x^2+1 ,y(0)=1:0.2:3
clc
clear all
close all
f=@(x,y) (y-x.^2+1)
[x,y]=ode45(f,[0,-2],1:0.2:3);
plot(x,y)
hold on
[x1,y1]=ode45(f,[0,2],1:0.2:3);
plot(x1,y1)
hold off
xlabel('x-axis')
ylabel('y-axis')
title('Family of approximation solutions of IVPs')
grid
```

L09_Q07: Find the numerical solution of system of non linear initial value problems.

$$\frac{dx}{dt} = -x + 3z, \quad \frac{dy}{dt} = -y + 2z, \quad \frac{dz}{dt} = x^2 - 2z$$

$x(0) = 1, y(0) = \frac{1}{2}, z(0) = 3, tspan = [0,3]$ by using ode45 command.

```
%Find the numerical solution of system of Non linear initial value problems.
%dx/dt=-x+3z,dy/dt=-y+2z,dz/dt=x^2-2z
%x(0)=1,y(0)=1/2,z(0)=3,tspan=[0,3] by using ode45 command
clc
clear all
close all
format short
tspan=[0,1.5];
x0=1;
y0=0.5;
z0=3;
f=@(t,x) [-x(1)+3*x(3);-x(2)+2*x(3);x(1)^2-2*x(3)];
[t,x]=ode45(f,tspan,[x0 y0 z0])
plot3(x(:,1),x(:,2),x(:,3))
xlabel('x-axis')
ylabel('y-axis')
title('Numerical Solution of IVP')
grid
```

L09_Q08: Find the numerical solution of system of initial value problems. $\frac{dy}{dt} = 2t, y(0) = 0, tspan = [0,5]$ by using ode45 command.

```
%Find the numerical solution of system of initial value problems.
%dy/dt=2t
%y(0)=0,tspan=[0,5] by using ode45 command
```



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```
clc
clear all
close all
f=@(t,y) (2*t);
tspan=[0,5];
y0=0;
[t,y]=ode45(f,tspan,y0)
plot(t,y)
xlabel('x-axis')
ylabel('y-axis')
title('Numerical Solution of IVP')
grid
```

L09_Q09: Compare the exact and approximate solution of $\frac{dy}{dt} = t + y$, $y(0) = 3$, $tspan = [0,5]$ by using ode45 command.

```
%Compare the exact and approximate solution of dy/dt=t+y ,
%y(0)=3,tspan=[0,5] by using ode45 command
clc
clear all
close all
format short
f=@(t,y) (t+y);
tspan=[0,5];
y0=3;
[t,y]=ode45(f,tspan,y0);
exactsoln=dsolve('Dy=t+y','y(0)=3','t');
exact=eval(vectorize(exactsoln));
table=[t,y,exact]
plot(t,y,'b',t,exact,'r')
legend('Approximate','Exact')
xlabel('x-axis')
ylabel('y-axis')
title('Comparing exact and approximate solution of ODE')
```

L10_Q10: Create ROW VECTOR by following four ways. Elements separated by commas, elements separated by spaces, by using colon when elements of row vector iterate with equal interval, by using colon sign and increment value by between starting and ending element.

```
%Create ROW VECTOR by following four ways.
% 1. Elements separated by commas.
% 2. Elements separated by spaces.
% 3. By using colon when elements of row vector iterate with equal interval.
% 4. By using colon sign & increment value by b/w starting & ending element.
clc
clear all
close all
echo on
RV1=[2,3,4,5,6]
RV2=[2 3 4 5 6]
RV3=2:6
RV4=2:1:6
echo off
```

L10_Q11: Create COLUMN VECTOR by following ways. Elements separated by semi colon, elements separated by spaces or commas and then taking transpose.

```
% Create COLUMN VECTOR by following ways.
% 1. Elements separated by semi colon
% 2. Elements separated by spaces or commas and then taking transpose.
clc
clear all
close all
echo on
```




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```
CV1=[2;3;4;5;6]
RV1=[2 3 4 5 6];
CV2=RV1'
RV2=[2 3 4 5 6];
CV3=RV2'
echo off
```

L10_Q12: Create a matrix of 2 or more rows and columns. Elements in each row are separated by spaces or commas while elements in each column are separated by semicolon.

```
% Create a matrix of 2 or more rows and columns.
% Elements in each row are separated by spaces or commas
% while elements in each column are separated by semicolon.
clc
clear all
close all
echo on
M2by5=[2,3,4,5,6;12,13,14,15,16]
M2by5=[2 3 4 5 6;12 13 14 15 16]
echo off
```

L10_Q13: Generate a symmetric matrix by using pascal built-in function.

```
% Generate a symmetric matrix by using pascal built-in function.
clc
clear all
close all
echo on
A=pascal(2)
AA=transpose(A)
echo off
```

L10_Q14: Generate a anti symmetric matrix by using magic built-in function.

```
% Generate a anti symmetric matrix by using magic built-in function.
clc
clear all
close all
echo on
A=magic(3)
AA=transpose(A)
echo off
```

L10_Q15: Generate a diagonal matrix by using a vector and diag built-in function.

```
% Generate a diagonal matrix by using a vector and diag built-in function.
clc
clear all
close all
echo on
A=[1,2,3,4]
B=diag(A)
echo off
```

L10_Q16: Generate a diagonal vector of a magic matrix by using diag built-in function.

```
% Generate a diagonal vector of a magic matrix by using diag built-in function.
clc
clear all
close all
echo on
A=magic(3)
AA=diag(A)
AA=diag(A')
echo off
```



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L10_Q17: Generate 4-by-4 matrix on ones by using built-in function 'ones'. Generate 4-by-4 identity matrix by using built-in function 'eye'. Generate 4-by-4 null matrix by using built-in function 'zero'.

```
% Generate 4-by-4 matrix on ones by using built-in function 'ones'.  
% Generate 4-by-4 identity matrix by using built-in function 'eye'.  
% Generate 4-by-4 null matrix by using built-in function 'zero'.  
clc  
clear all  
close all  
echo on  
A=ones(4)  
A=eye(4)  
A=zeros(4)  
echo off
```

L10_Q18: Add and subtract scalar to each element of a vector.

```
% Add and subtract scalar to each element of a vector.  
clc  
clear all  
close all  
echo on  
A=[1 2 3 4]  
AA=A+3  
AA=A-3  
echo off
```

L10_Q19: Addition of two matrixes with same order.

```
% Addition of two matrixes with same order.  
clc  
clear all  
close all  
echo on  
A=[1 2 3 4];  
B=[3 5 6 7];  
C=A+B  
D=A-B  
echo off
```

L10_Q20: Find determinant of a matrix.

```
% Find determinant of a matrix.  
clc  
clear all  
close all  
echo on  
A=[1 2 3;4 5 6;5 2 1];  
B=det(A)  
echo off
```

L10_Q21: Find inverse matrix of a matrix.

```
% Find inverse matrix of a matrix.  
clc  
clear all  
close all  
format short  
echo on  
A=[1 2 3;4 5 6;5 2 1];  
B=inv(A)  
echo off
```

L10_Q22: Multiplication of two matrixes by using for loop.



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% Multiplication of two matrixes by using for loop.

```
clc
clear all
close all
format short
echo on
A=[1 2 3;4 5 6];
%order of A=2 by 3
B=[3 4 6;4 5 1;4 3 1];
%order of B=3 by 3
for i=1:2
    for j=1:3
        C(i,j)=A(i,:)*B(:,j)
    end
end
C
C1=A*B
echo off
```

L10_Q23: By using sum command compute a vector of row elements.

% By using sum command compute a vector of row elements.

```
clc
clear all
close all
echo on
A=sym([3 4/3 6;4 5 1;4 3 1]);
B=sum(A)
A=[3 4/3 6;4 5 1;4 3 1];
B=sum(A)
echo off
```

L10_Q24: By using sum command compute a vector of column elements.

% By using sum command compute a vector of column elements.

```
clc
clear all
close all
echo on
A=sym([3 4 6;4 5 1;4 3 1]);
B=sum(A')
A=[3 4 6;4 5 1;4 3 1];
B=sum(A')
echo off
```

L10_Q25: By using sum and end command compute a vector of row elements.

% By using sum and end command compute a vector of row elements.

```
clc
clear all
close all
echo on
A=sym([3 4 6;4 5 1;4 3 1]);
B=sum(A(:,1))
B=sum(A(:,2))
B=sum(A(:,end))
echo off
```

L10_Q26: Multiplication of two matrixes by using for loop.

% Multiplication of two matrixes by using for loop.

```
clc
clear all
```



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```
close all
format short
echo on
A=[1;2;3];
%order of A=3 by 1
B=[3 5 1];
%order of B=1 by 3
for i=1:3
    for j=1:3
        C(i,j)=A(i,:)*B(:,j)
    end
end
C
C1=A*B
echo off
```

L10_Q27: Multiplication of two matrixes by using for loop.

```
% Multiplication of two matrixes by using for loop.
clc
clear all
close all
format short
echo on
A=sym([5/6,1/6,0;1/6,0,5/6;0,5/6,1]);
%order of A=3 by 3
B=sym([5/2,1/2,0;1/3,0,5/3;0,5/6,1]);
%order of B=3 by 3
for i=1:3
    for j=1:3
        C(i,j)=A(i,:)*B(:,j)
    end
end
C
C1=A*B
echo off
```

L10_Q28: Find the sum of second column or second row of a matrix.

```
% Find the sum of second column or second row of a matrix.
clc
clear all
close all
echo on
A=[2 4 6;3 1 9;2 4 6];
SumCol2=sum(A(:,2))
SumRow1=sum(A(2,:))
echo off
```

L11_Q29: Compute the Eigen values and Eigen vectors of the following matrix

$$B = \begin{bmatrix} 0 & 1 & 3 \\ 2 & 4 & 1 \\ 6 & 1 & 8 \end{bmatrix}$$

```
% Compute the Eigen values and Eigen vectors of the following matrix
clc
clear all
close all
echo on
A=[0,1,3;2,4,1;6,1,8]
eig(A)
[V,D]=eig(A)
echo off
```




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L11_Q30: Find reduced row echelon form of the following matrix

$$B = \begin{bmatrix} 0 & 1 & 3 \\ 2 & 4 & 1 \\ 6 & 1 & 8 \end{bmatrix}$$

```
% Find reduced row echelon form of the following matrix
clc
clear all
close all
echo on
A=[0,1,3;2,4,1;6,1,8]
RREFA=rref(A)
echo off
```

L11_Q31: Find solution of the following system of linear equations.

$$x_1 + 3x_2 = -2, \quad 2x_1 + 4x_2 = 1$$

```
% Find solution of the following system of linear equations.
% x1+3x2=-2; 2x1+4x2=1
clc
clear all
close all
echo on
CoefmA=sym([1,3;2,4])
ColmB=sym([-2;1])
C=inv(CoefmA);
X=C*ColmB
echo off
```

L11_Q32: Solve following system of linear equations by using **solve** command.

$$3x + 2y + z = 1, \quad y = 2, \quad x + 2y = 3$$

```
% Solve following system of linear equations by using solve command.
%3x+2y+z=1; y=2; x+2y=3
clc
clear all
close all
echo on
eq1='3*x+2*y+z=1'
eq2='y=2'
eq3='x+2*y=3'
soln=solve(eq1,eq2,eq3)
X=double([soln.x,soln.y,soln.z])
echo off
```

L11_Q33: Solve following system of linear equations by following four methods

$$x + 2y + 4z = 1, \quad 3x + 2y + 4z = 2, \quad 2x + 4y + 5z = 5$$

By 'integer division', By 'inverse if possible', By using 'solve command' and by using 'mldivide command'.

```
% Solve following system of linear equations by following four methods
%x+2y+4z=1; 3x+2y+4z=2; 2x+4y+5z=5
% 1. By 'integer division', 2. By 'inverse if possible',
% 3. By using 'solve command' 4. By using 'mldivide command'.
clc
clear all
close all
echo on
A=[1,2,4;3,2,4;2,4,5]
B=[1;2;5]
% 1. By 'integer division'.
X=A\B
```




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```
%2. By 'inverse if possible',  
C=inv(A);  
X=C*B  
% 3. By using 'solve command'  
eq1='x+2*y+4*z=1';  
eq2='3*x+2*y+4*z=2';  
eq3='2*x+4*y+5*z=5';  
soln=solve(eq1,eq2,eq3);  
X=double([soln.x,soln.y,soln.z])'  
%4. By using 'mldivide command'.  
X=mldivide(A,B)  
echo off
```

L11_Q34: Consider the system of linear equations;

$$3x + 2y + z = 1, \quad y = 2, \quad x + 2y = 3$$

Enter it into MATLAB as an augmented matrix named Ab; Use elementary row operations to reduced echelon form.

```
% Consider the system of linear equations;  
% 3x+2y+z=1; y=2 ; x+ 2y=3  
% Enter it into MATLAB as an augmented matrix named Ab;  
% Use elementary row operations to reduced echelon form.  
clc  
close all  
clear all  
echo on  
format short  
A=[3,2,1;0,1,0;1,2,0];  
b=[1,2,3]';  
Ab=[A,b];  
Ab(1,:)=1/3*Ab(1,:);  
Ab(3,:)=Ab(3,:)-Ab(1,:);  
Ab(1,:)=Ab(1,:)-0.6667*Ab(2,:);  
Ab(3,:)=Ab(3,:)-1.3333*Ab(2,:);  
Ab(3,:)=1/-0.3333*Ab(3,:);  
Ab(1,:)=Ab(1,:)-0.3333*Ab(3,:);  
X=Ab(:,4)
```

L11_Q35: Show that $\begin{pmatrix} a \\ b \end{pmatrix}$ is in the span of $\begin{pmatrix} -2 & 1 \\ 1 & 6 \end{pmatrix}$.

```
% Show that [a;b] is in the span of [-2,1;1,6]  
clc  
close all  
clear all  
echo on  
format short  
syms a b  
A=[-2,1;1,6];  
b=[a;b];  
Ab=[A b];  
sol=rref(Ab)  
disp('Possible values of weights');  
X=sol(:,3)
```

L11_Q36: Consider the system of linear equations;

$$x_1 + 2x_2 - 3x_3 = -3, \quad -4x_1 - 5x_2 + 2x_3 = -2, \quad 2x_1 + 3x_2 - x_3 = 2$$

Enter it into MATLAB as an augmented matrix named Ab; Use elementary row operations to reduced echelon form. Give the solution of the system. (If the system has no solution say that. Otherwise write the basic variables (Those corresponding to pivot columns) as function of numbers and free variables, if there are any).

```
% Consider the system of linear equations;
```



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```
%x1+2*x2-3*x3=-3
%-4*x1-5*x2+2*x3=-2
%2*x1+3*x2-x3=2
%Enter it into MATLAB as an augmented matrix named Ab;
%Use elementary row operations to reduced echelon form.
%Give the solution of the system.
clc
close all
clear all
echo on
format short
A=[1,2,-3;-4,-5,2;2,3,-1];
b=[-3;-2;2];
Ab=[A b]
Ab(2,:)=Ab(2,:)+4*Ab(1,:);
Ab(3,:)=Ab(3,:)-2*Ab(1,:);
Ab(2,:)=1/3*Ab(2,:);
Ab(3,:)=Ab(2,:)+Ab(3,:);
Ab(1,:)=2*Ab(2,:)-Ab(1,:);
Ab(1,:)=(-1)*Ab(1,:);
Ab(3,:)=(1/1.6667)*(Ab(3,:));
Ab(2,:)=(3.3333)*(Ab(3,:))+(Ab(2,:));
Ab(1,:)=(3.6667)*(Ab(3,:))-(Ab(1,:));
Ab(1,:)=(-1)*Ab(1,:);
X=Ab(:,4)
```

L11_Q37: Consider the system of linear equations;

$$x_1 + 3x_3 + x_5 = -1, \quad 4x_1 + x_2 + x_3 + 6x_5 = 1, \quad -3x_1 - 3x_2 - 3x_3 + x_4 - 19x_5 = 6, \\ 10x_1 - 4x_2 + 38x_3 + 2x_4 - 12x_5 = 0$$

Enter it into MATLAB as an augmented matrix named Ab; Use elementary row operations to reduced echelon form. Give the solution of the system. (If the system has no solution say that. Otherwise write the basic variables (Those corresponding to pivot columns) as function of numbers and free variables, if there are any).

```
% Consider the system of linear equations;
% x1+3*x3+x5=-1
% 4*x1+x2+x3+6*x5=1
% -3*x1-3*x2-3*x3+x4-19*x5=6
% 10*x1-4*x2+38*x3+2*x4-12*x5=0
%Enter it into MATLAB as an augmented matrix named Ab;
%Use elementary row operations to reduced echelon form.
%Give the solution of the system.
clc
close all; clear all
echo on
format short
A=[1,0,3,0,1;4 1 1 0 6;-3 -3 -3 1 -19;10 -4 38 2 -12];
b=[-1;1;6;0];
Ab=[A b]
Ab(2,:)=4*Ab(1,:)-Ab(2,:);
Ab(3,:)=3*Ab(1,:)+Ab(3,:);
Ab(4,:)=10*Ab(1,:)-Ab(4,:);
Ab(3,:)=3*Ab(2,:)-Ab(3,:);
Ab(4,:)=4*Ab(2,:)+Ab(4,:);
Ab(2,:)=(-1)*Ab(2,:);
Ab(3,:)=(1/27)*Ab(3,:);
Ab(4,:)=36*Ab(3,:)-Ab(4,:);
Ab(2,:)=11*Ab(3,:)+Ab(2,:);
Ab(1,:)=3*Ab(3,:)-Ab(1,:);
Ab(4,:)=(1/0.6667)*Ab(4,:);
Ab(1,:)=(0.1111)*Ab(4,:)+Ab(1,:);
```



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```
Ab(2,:)=(0.4074)*Ab(4,:)+Ab(2,:);
Ab(3,:)=(0.0370)*Ab(4,:)+Ab(3,:);
Ab(1,:)=(-1)*Ab(1,:);
Ab
disp('x5 is free variable - Let x5=1 then other values will be as under')
x=Ab(:,6)-Ab(:,5)
x1=1
x2=x(1,:)
x3=x(2,:)
x4=x(3,:)
x5=x(4,:)
```

L11_Q38: Show that $\begin{pmatrix} h \\ k \end{pmatrix}$ is in the span of $\begin{pmatrix} -3 & 2 \\ 4 & 5 \end{pmatrix}$.

```
% Show that [h;k] is in the span of [-3,2;4,5]
clc
close all
clear all
echo on
format short
syms h k
A=[-3,2;4,5];
b=[h;k];
Ab=[A b];
sol=rref(Ab)
disp('Possible values of weights');
X=sol(:,3)
```

L11_Q39: Solve the matrix equations of Sylvester type i.e. $AX+XB=C$. The matrices A,B and C are known and Matrix X to be determined.

```
% Solve the matrix equations of Sylvester type i.e. AX+XB=C.
% The matrices A,B and C are known and Matrix X to be determined.
clc
clear all
close all
echo on
A=[2,3,4;4,5,2;2,3,5]
B=[-3,3,4;-8,-3,2;0,3,5]
C=[-6,0,4;3,5,9;2,3,4]
X=sylvester(A,B,C)
echo off
```

L12&15_Q40: Compute the Riemann sum of the function $f(x) = e^{-x^2}$ for $n=4, 6, 8, 10, 100$ and 1000 by choosing C_k as the right end point of the interval $[0, 2]$.

```
% Compute the Riemann sum of the function f(x)=e^(-x^2)
% for n=4, 6, 8, 10, 100 and 1000
% by choosing CK as the right end point of the interval [0, 2].
clc
clear all;
close all;
a=0;
b=2;
n=100;
%ALSO CHECK RESULTS BY PUTTING n=4, n=6, n=8, n=10 and n=1000
value=0;
f=inline('exp(-x^2)');
dx=(b-a)/n;
for k=1:n
    c=a+k*dx;
    value=value+f(c);
end
value=dx*value
```




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L12_Q41: Compute the Riemann sum of the function $f(x) = e^{-x^2}$ for $n=100$ by choosing C_K is randomly selected on the interval $[0, 2]$.

```
% Compute the Riemann sum of the function f(x)=e^(-x^2) for n=100  
% by choosing CK is randomly selected on the interval [0, 2].
```

```
clear all;  
close all;  
clc;  
a=0;  
b=2;  
n=100;  
value=0;  
f=inline('exp(-x^2)');  
dx=(a-b)/n;  
for k=1:n  
    c=dx*rand+(a+(k-1)*dx)  
    value=value+f(c);  
end  
value=dx*value;
```

L12_Q42: Find the Riemann sum of the function $f(x) = x^2$ for $n=100$ by choosing C_K is randomly selected on the interval $[-1, 3]$.

```
% Find the Riemann sum of the function f(x)=x^2 for n=100  
% by choosing CK is randomly selected on the interval [-1, 3].
```

```
clear all;  
close all;  
clc;  
a=-1;  
b=3;  
n=100;  
value=0;  
f=inline('x^2');  
dx=(a-b)/n;  
for k=1:n  
    c=dx*rand+(a+(k-1)*dx)  
    value=value+f(c);  
end  
value=dx*value;
```

L12&15_Q43: For the integral $\int_1^5 \sqrt{1+x^3} dx$ compute the Riemann sum approximation using 100 equally spaced sub-intervals and left end points.

```
% For the integral sqrt((1+x^3))dx [1, 5]  
% compute the Riemann sum approximation  
% using 100 equally spaced sub-intervals and left end points.
```

```
function value=riemannsum(f,a,b,n)  
clear all;  
close all;  
clc;  
a=1;  
b=5;  
n=100;  
value=0;  
f=@(x) (sqrt(1+x.^3));  
dx=(b-a)/n;  
for k=1:n  
    c=a+(k-1)*dx;  
    value=value+f(c);  
end  
value=dx*value;
```

L12&15_Q44: For the integral $\int_1^5 \sqrt{1+x^3} dx$ compute the Riemann sum approximation using 100 equally spaced sub-intervals and right end points.

```
% For the integral sqrt((1+x^3))dx [1, 5]
```



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```
% compute the Riemann sum approximation
% using 100 equally spaced sub-intervals and right end points.
function value=riemannsum(f,a,b,n)
clear all;
close all;
clc;
a=1;
b=5;
n=100;
value=0;
f=@(x) (sqrt(1+x.^3));
dx=(b-a)/n;
for k=1:n
    c=a+(k)*dx;
    value=value+f(c);
end
value=dx*value;
```

L12&15_Q45: For the integral $\int_1^5 \sqrt{1+x^3} dx$ compute the Riemann sum approximation using 100 equally spaced sub-intervals and mid points.

```
% For the integral sqrt((1+x^3))dx [1, 5]
% compute the Riemann sum approximation
% using 100 equally spaced sub-intervals and right end points.
function value=riemannsum(f,a,b,n)
clear all;
close all;
clc;
a=1;
b=5;
n=100;
value=0;
f=@(x) (sqrt(1+x.^3));
dx=(b-a)/n;
for k=1:n
    c=((a+(k)*dx)+(a+(k-1)*dx))/2;
    value=value+f(c);
end
value=dx*value;
```

L12&15_Q46: Compute the definite integral of the function $f(x) = e^{-x^2}$ using Trapezoid Rule considering 100 sub-intervals (that is, $n=100$) on the interval $[0, 2]$.

```
% Compute the definite integral of the function f(x)=e^(-x^2) using Trapezoid Rule
% considering 100 sub-intervals (that is, n=100) on the interval [0, 2].
clear all;
clc;
close all;
sum=0;
a=0;
b=2;
n=100;
f=inline('exp(-x^2)');
dx=(b-a)/n;
Averagevalue=(f(b)+f(a))/2;
for k=1:(n-1)
    c = a+k*dx;
    sum=sum+f(c);
end
Answer=(sum+Averagevalue)*dx
```

L12_Q47: Compute the definite integral of the function $f(x) = \sin(x)$ using Trapezoid Rule on the interval $[0, \pi]$ for $n=8, 12, 100$.

```
% Compute the definite integral of the function f(x)=sin(x)
% using Trapezoid Rule on the interval [0, pi] for n=8, 12, 100.
```



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```
close all
clear all
clc
a=0;
b=pi;
n=8;
%ALSO CHECK RESULTS BY PUTTING n=12 and n=100
dx=(b-a)/n;
f=@(x)(sin(x));
AW=(f(a)+f(b))/2;
for k=1:n-1
    x(k)=a+k*dx;
end
funvalue=f(x);
answer=(sum(funvalue)+AW)*dx
```

L12&15_Q48: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Trapezoid Rule considering 10 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [1, 5]
% using Trapezoid Rule considering 10 equally spaced sub-intervals.
close all
clear all
clc
a=1;
b=5;
n=10;
dx=(b-a)/n;
f=@(x)(sqrt(1+x.^3));
AW=(f(a)+f(b))/2;
for k=1:n-1
    x(k)=a+k*dx;
end
funvalue=f(x);
answer=(sum(funvalue)+AW)*dx
```

L12&15_Q49: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Trapezoid Rule considering 100 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [1, 5]
% using Trapezoid Rule considering 100 equally spaced sub-intervals.
close all
clear all
clc
a=1;
b=5;
n=100;
dx=(b-a)/n;
f=@(x)(sqrt(1+x.^3));
AW=(f(a)+f(b))/2;
for k=1:n-1
    x(k)=a+k*dx;
end
funvalue=f(x);
answer=(sum(funvalue)+AW)*dx
```

L12&15_Q50: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Trapezoid Rule considering 1000 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [1, 5]
% using Trapezoid Rule considering 1000 equally spaced sub-intervals.
close all
clear all
clc
a=1;
```




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```
b=5;  
n=1000;  
dx=(b-a)/n;  
f=@(x)(sqrt(1+x.^3));  
AW=(f(a)+f(b))/2;  
for k=1:n-1  
    x(k)=a+k*dx;  
end  
funvalue=f(x);  
answer=(sum(funvalue)+AW)*dx
```

L12&15_Q51: Compute the definite integral of the function $f(x) = e^{-x^2}$ on the interval $[0, 2]$ using Simpson's one-third Rule by dividing the interval $[0, 2]$ into 6 sub-intervals.

%Compute the definite integral of f(x)=e^-x2 on the interval [0,2] using
%Simpson's one Third Rule by dividing interval [0,2] into 6 sub-intervals.

```
close all  
clear all  
clc  
a=0;  
b=2;  
n=6;  
dx=(b-a)/n  
f=@(x)exp(-x.^2)  
value1=0;  
value2=0;  
for k=1:n-1  
    x(k)=a+k*dx;  
    if mod(k,2)==0  
        value1=value1+2*f(x(k));  
    else  
        value2=value2+4*f(x(k));  
    end  
end  
answer=(value1+value2+f(a)+f(b))*(dx/3)
```

L12&15_Q52: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Simpson's One-Third Rule considering 6 equally spaced sub-intervals.

% Compute the following integral sqrt((1+x^3))dx [1, 5]
%Simpson's one Third Rule by considering 6 equally spaced sub-intervals.

```
close all  
clear all  
clc  
a=1;  
b=5;  
n=6;  
dx=(b-a)/n  
f=@(x)(sqrt(1+x.^3));  
value1=0;  
value2=0;  
for k=1:n-1  
    x(k)=a+k*dx;  
    if mod(k,2)==0  
        value1=value1+2*f(x(k));  
    else  
        value2=value2+4*f(x(k));  
    end  
end  
answer=(value1+value2+f(a)+f(b))*(dx/3)
```

L12&15_Q53: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Simpson's One-Third Rule considering 30 equally spaced sub-intervals.



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```
% Compute the following integral sqrt((1+x^3))dx [1, 5]
%Simpson's one Third Rule by considering 30 equally spaced sub-intervals.
close all
clear all
clc
a=1;
b=5;
n=30;
dx=(b-a)/n
f=@(x) (sqrt(1+x.^3));
value1=0;
value2=0;
for k=1:n-1
    x(k)=a+k*dx;
    if mod(k,2)==0
        value1=value1+2*f(x(k));
    else
        value2=value2+4*f(x(k));
    end
end
answer=(value1+value2+f(a)+f(b))*(dx/3)
```

L12&15_Q54: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Simpson's One-Third Rule considering 300 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [1, 5]
%Simpson's one Third Rule by considering 300 equally spaced sub-intervals.
close all
clear all
clc
a=1;
b=5;
n=300;
dx=(b-a)/n
f=@(x) (sqrt(1+x.^3));
value1=0;
value2=0;
for k=1:n-1
    x(k)=a+k*dx;
    if mod(k,2)==0
        value1=value1+2*f(x(k));
    else
        value2=value2+4*f(x(k));
    end
end
answer=(value1+value2+f(a)+f(b))*(dx/3)
```

L12_Q55: Compute the definite integral of the function $f(x) = x^4$ on the interval $[-3, 3]$ using Simpson's one-third Rule by taking $n=6, 12, 100$.

```
% Compute the following integral of function f(x)=x^4 on interval [-3, 3]
% using Simpson's one Third Rule by taking n=6, 12, 100.
close all
clear all
clc
a=-3;
b=3;
n=6;
%ALSO CHECK RESULTS BY PUTTING n=12 and n=100
dx=(b-a)/n
f=@(x) (x^4);
value1=0;
value2=0;
for k=1:n-1
```



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```
x(k)=a+k*dx;  
if mod(k,2)==0  
    value1=value1+2*f(x(k));  
else  
    value2=value2+4*f(x(k));  
end  
end  
answer=(value1+value2+f(a)+f(b))*(dx/3)
```

L12&15_Q56: Compute the definite integral of the function $f(x) = e^{-x^2}$ on the interval $[0, 2]$ using Simpson's Three-Eight Rule by dividing the interval $[0, 2]$ into 6 sub-intervals.

```
% Compute the following integral f(x)=e^(-x^2) on interval [0, 2]  
% using Simpson's Three-Eight Rule by considering 6 equally spaced sub-intervals.  
clear all;  
clc;  
close all;  
sum1=0;  
sum2=0;  
a=0;  
b=2;  
n=6;  
f=@(x) exp(-x.^2)  
dx=(b-a)/n;  
value=f(b)+f(a);  
for k=1:n-1  
    c=a+k*dx;  
    if mod(k,3)==0  
        sum1=sum1+2*f(c);  
    else  
        sum2=sum2+3*f(c);  
    end  
end  
sum=sum1+sum2;  
Answer=(value+sum)*(3*dx/8)
```

L12&15_Q57: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Simpson's Three-Eight Rule considering 6 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [1, 5]  
% using Simpson's Three-Eight Rule by considering 6 equally spaced sub-intervals.  
clear all;  
clc;  
close all;  
sum1=0;  
sum2=0;  
a=1;  
b=5;  
n=6;  
f=@(x) sqrt(1+x^3)  
dx=(b-a)/n;  
value=f(b)+f(a);  
for k=1:n-1  
    c=a+k*dx;  
    if mod(k,3)==0  
        sum1=sum1+2*f(c);  
    else  
        sum2=sum2+3*f(c);  
    end  
end  
sum=sum1+sum2;  
Answer=(value+sum)*(3*dx/8)
```




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L12&15_Q58: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Simpson's Three-Eight Rule considering 30 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [1, 5]
% using Simpson's Three-Eight Rule by considering 30 equally spaced sub-intervals.
clear all;
clc;
close all;
sum1=0;
sum2=0;
a=1;
b=5;
n=30;
f=@(x) sqrt(1+x^3)
dx=(b-a)/n;
value=f(b)+f(a);
for k=1:n-1
    c=a+k*dx;
    if mod(k,3)==0
        sum1=sum1+2*f(c);
    else
        sum2=sum2+3*f(c);
    end
end
sum=sum1+sum2;
Answer=(value+sum)*(3*dx/8)
```

L12&15_Q59: Compute the following integral $\int_1^5 \sqrt{1+x^3} dx$ using Simpson's Three-Eight Rule considering 300 equally spaced sub-intervals.

```
% Compute the following integral sqrt((1+x^3))dx [1, 5]
% using Simpson's Three-Eight Rule by considering 300 equally spaced sub-intervals.
clear all;
clc;
close all;
sum1=0;
sum2=0;
a=1;
b=5;
n=300;
f=@(x) sqrt(1+x^3)
dx=(b-a)/n;
value=f(b)+f(a);
for k=1:n-1
    c=a+k*dx;
    if mod(k,3)==0
        sum1=sum1+2*f(c);
    else
        sum2=sum2+3*f(c);
    end
end
sum=sum1+sum2;
Answer=(value+sum)*(3*dx/8)
```

L12&15_Q60: Compute the Riemann sum of the function $f(x) = e^{-x^2}$ for $n=4, 6, 8, 10, 100$ and 1000 by choosing C_k as the left end point of the interval $[0, 2]$.

```
% Compute the Riemann sum of the function f(x)=e^(-x^2)
% for n=4, 6, 8, 10, 100 and 1000 by choosing CK as
% the left end point of the interval [0, 2].
clear all
close all
clc
```



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```
a=2;
b=0;
n=4;
%ALSO CHECK RESULTS BY PUTTING n=6, n=8, n=10, n=100 and n=1000
dx=(b-a)/n
f=@(x)exp(-x.^2)
for k=1:n
    c(k)=a+k*dx;
end
funcvalue=f(c);
summation=sum(funcvalue);
answer=summation*dx
```

L12&15_Q61: Compute the Riemann sum of the function $f(x) = e^{-x^2}$ for $n=4, 6, 8, 10, 100$ and 1000 by choosing C_K as the right end point of the interval $[0, 2]$.

```
% Compute the Riemann sum of the following function f(x)=e^-x^2 for
% n=4,6,8,10,100,1000 by choosing ck as the right point of interval[0,2]
clear all
close all
clc
a=0;
b=2;
n=4;
%ALSO CHECK RESULTS BY PUTTING n=6, n=8, n=10, n=100 and n=1000
dx=(b-a)/n
f=@(x)exp(-x.^2)
for k=1:n
    c(k)=a+k*dx;
end
funcvalue=f(c);
summation=sum(funcvalue);
answer=summation*dx
```

L12&15_Q62: Compute the Riemann sum of the function $f(x) = e^{-x^2}$ for $n=4, 6, 8, 10, 100$ and 1000 by choosing C_K as the right mid point of the interval $[0, 2]$.

```
% Compute the Riemann sum of the following function f(x)=e^-x^2 for
% n=4,6,8,10,100,1000 by choosing ck as the mid point of interval[0,2]
clear all
close all
clc
a=0;
b=2;
n=4;
%ALSO CHECK RESULTS BY PUTTING n=6, n=8, n=10, n=100 and n=1000
dx=(b-a)/n
f=@(x)exp(-x.^2)
for k=1:n
    c=dx*rand+(a+(k-1)*dx);
end
funcvalue=f(c);
summation=sum(funcvalue);
answer=summation*dx
```

L13_Q63: Construct the function vector $p_n^{(\alpha, \beta)}(x)$ (Orthogonal Shifted Jacobi Polynomials) defined on $[0, T]$ of degree 0, 1, 2, and 3 using equation

$$p_n^{(\alpha, \beta)}(x) = \sum_{k=0}^n \frac{(-1)^{(i-k)} \Gamma(i + \beta + 1) \Gamma(i + k + \alpha + \beta + 1)}{\Gamma(k + \beta + 1) \Gamma(i + \alpha + \beta + 1) \Gamma(i - k + 1) \Gamma(k + 1) T^k} x^k \text{ where } i = 0, 1, 2, 3, \dots$$

```
% Construct the function vector (Orthogonal Shifted Jacobi Polynomials)
% defined on [0,T] by alpha=0 and beta=0 and scale level M=4 taking using equation
function R=JacobFunVec(alpha,beta,M)
```



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```
clear all
close all
clc
syms x T k
M=4;
alpha=0;
beta=0;
for i=0:M-1
    a1=(-1)^(i-k);
    a2=gamma(i+beta+1);
    a3=gamma(i+k+alpha+beta+1);
    d1=gamma(k+beta+1);
    d2=gamma(i+alpha+beta+1);
    d3=gamma(i-k+1);
    d4=gamma(k+1);
    d5=T^k;
    aa=(a1*a2*a3)/(d1*d2*d3*d4*d5);
    P(i+1)=symsum(aa*x^k,k,0,i);
end
R=transpose(P)
```

L13_Q64: Construct the function vector $p_n^{(\alpha,\beta)}(x)$ (Orthogonal Shifted Jacobi Polynomials) defined on $[0,1]$ by $\alpha = 1$ and $\beta = 2$ and scale level $M=10$ taking using equation

$$p_n^{(\alpha,\beta)}(x) = \sum_{k=0}^n \frac{(-1)^{(i-k)} \Gamma(i+\beta+1) \Gamma(i+k+\alpha+\beta+1)}{\Gamma(k+\beta+1) \Gamma(i+\alpha+\beta+1) \Gamma(i-k+1) \Gamma(k+1) T^k} x^k \text{ where } i = 0, 1, 2, 3, \dots$$

% Construct the function vector (Orthogonal Shifted Jacobi Polynomials)
% defined on $[0,1]$ by $\alpha=1$ and $\beta=2$ and scale level $M=10$ taking using equation

```
function P=JacobFunVec(alpha,beta,M)
clear all
close all
clc
syms x T k
M=10;
T=1;
alpha=1;
beta=2;
for i=0:M-1
    a1=(-1)^(i-k);
    a2=gamma(i+beta+1);
    a3=gamma(i+k+alpha+beta+1);
    d1=gamma(k+beta+1);
    d2=gamma(i+alpha+beta+1);
    d3=gamma(i-k+1);
    d4=gamma(k+1);
    d5=T^k;
    aa=(a1*a2*a3)/(d1*d2*d3*d4*d5);
    P(i+1)=symsum(aa*x^k,k,0,i);
end
P=transpose(P);
```

L13_Q65: Write down a MATLAB code for weight function i.e. $W_T^{(\alpha,\beta)}(x) = (T-X)^\alpha x^\beta$

% Write down a MATLAB code for weight function

```
clc
clear all
close all
syms T x alpha beta
W=(T-x)^alpha*(x^beta)
```

L13_Q66: Construct the weight function vector $W_T^{(\alpha,\beta)}(x) = (T-X)^\alpha x^\beta$ on the interval $[0,1]$ by $\alpha = 1$ and $\beta = 2$.



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% Construct the weight function vector on the interval [0,1] by $\alpha=1$ and $\beta=2$.

```
clc
clear all
close all
syms T x alpha beta
alpha=1;
beta=2;
T=1;
W=(T-x)^alpha*(x^beta)
```

L13_Q67: Construct the weight function vector $W_T^{(\alpha,\beta)}(x) = (T-X)^\alpha x^\beta$ on the interval [0,1] by $\alpha = 0$ and $\beta = 0$.

% Construct the weight function vector on the interval [0,1] by $\alpha=0$ and $\beta=0$.

```
clc
clear all
close all
syms T x alpha beta
alpha=0;
beta=0;
T=1;
W=(T-x)^alpha*(x^beta)
```

L13_Q68: Check the orthogonality of orthogonal shifted Jacobi polynomial $p_n^{(\alpha,\beta)}(x)$ on [0,T] by choosing $\alpha = 1$ and $\beta = 1$ with respect to the weight function $W_T^{(\alpha,\beta)}(x)$.

SORRY

L13_Q69: Write down a MATLAB code for generating the constant factor

$$R_{T,j}^{(\alpha,\beta)} \delta_{(i,j)} = \frac{T^{\alpha+\beta+1} \Gamma(j+\alpha+1) \Gamma(j+\beta+1)}{(2j+\alpha+\beta+1) \Gamma(j+1) \Gamma(j+\alpha+\beta+1)}$$

% Write down a MATLAB code for generating the constant factor

```
function R=ConstantFactor(alpha,beta,T,M)
clear all
close all
clc
syms j alpha beta T
T=1;
alpha=1;
beta=1;
M=4;
for j=0:M-1
    a1=T^(alpha+beta+1);
    a2=gamma(j+alpha+1);
    a3=gamma(j+beta+1);
    d1=(2*j+alpha+beta+1);
    d2=gamma(j+1);
    d3=gamma(j+alpha+beta+1);
    R(j+1)=(a1*a2*a3)/(d1*d2*d3);
end
R=transpose(R)
```

L13_Q70: Compute the constant factor defined in $R_{T,j}^{(\alpha,\beta)} \delta_{(i,j)} = \frac{T^{\alpha+\beta+1} \Gamma(j+\alpha+1) \Gamma(j+\beta+1)}{(2j+\alpha+\beta+1) \Gamma(j+1) \Gamma(j+\alpha+\beta+1)}$ for $j=0:5$ and $\alpha = 2$ and $\beta = 3$.

% Compute the constant factor defined for $j=0:5$ and $\alpha=2$ and $\beta=3$.

```
function R=ConstantFactor(alpha,beta,T)
clear all
close all
clc
syms j alpha beta T
T=1;
```



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```
alpha=2;
beta=3;
for j=0:5
    a1=T^(alpha+beta+1);
    a2=gamma(j+alpha+1);
    a3=gamma(j+beta+1);
    d1=(2*j+alpha+beta+1);
    d2=gamma(j+1);
    d3=gamma(j+alpha+beta+1);
    R(j+1)=(a1*a2*a3)/(d1*d2*d3);
end
R=transpose(R)
```

L13_Q71: Compute the constant factor defined in $R_{T,j}^{(\alpha,\beta)} \delta_{(i,j)} = \frac{T^{\alpha+\beta+1} \Gamma(j+\alpha+1) \Gamma(j+\beta+1)}{(2j+\alpha+\beta+1) \Gamma(j+1) \Gamma(j+\alpha+\beta+1)}$ for choosing the scale level $M=7$ and $\alpha = 3$ and $\beta = 1$.

% Compute the constant factor defined for choosing the scale level $M=7$ and $\alpha=3$ and $\beta=1$.

```
function R=ConstantFactor(alpha,beta,T,M)
clear all
close all
clc
syms j alpha beta T
T=1;
alpha=3;
beta=1;
M=7;
for j=0:M-1
    a1=T^(alpha+beta+1);
    a2=gamma(j+alpha+1);
    a3=gamma(j+beta+1);
    d1=(2*j+alpha+beta+1);
    d2=gamma(j+1);
    d3=gamma(j+alpha+beta+1);
    R(j+1)=(a1*a2*a3)/(d1*d2*d3);
end
R=transpose(R)
```

L13_Q72: Write down a MATLAB code for computing the series coefficients C_i , $i=0,1,2,3,4$.

$$C_i = \frac{1}{R_{T,j}^{(\alpha,\beta)}} \int_0^T f(x) W_T^{(\alpha,\beta)}(x) P_i^{(\alpha,\beta)}(x) dx$$

% Write down a MATLAB code for computing the series coefficients C_i , $i=0,1,2,3,4$.

```
function C=JacobiCoeffVector(f,x,M,alpha,beta,T)
close all;
clear all;
clc;
syms f x M alpha beta T k i j
T=1;
alpha=1;
beta=1;
M=4;
W = (T - x).^alpha * x.^beta;
for i=0:M-1
    a1 = (-1)^(i-k);
    a2 = gamma(i + beta+1);
    a3 = gamma(i + k+ alpha + beta + 1);
    d1 = gamma(k+beta + 1);
    d2 = gamma(i + alpha + beta + 1);
    d3 = gamma(i - k + 1);
    d4 = gamma(k+1);
    d5 = T^k;
    aa = (a1*a2*a3)/(d1*d2*d3*d4*d5);
```



```
P(i+1) = symsum(aa*x^k, k, 0, i);
end
P;
PW=f*P*W;
for j = 0:M-1
    a1 = T^(alpha + beta +1);
    a2 = gamma(j + alpha + 1);
    a3 = gamma(j + beta + 1);
    d1 = 2*j + alpha + beta + 1;
    d2 = gamma(j + 1);
    d3 = gamma(j + alpha + beta + 1);
    R(j + 1) = (a1*a2*a3)/(d1*d2*d3);
end
R ;
RR=1./R;
for i = 0:M-1
    C(i+1)=int(PW(i+1),x,0,i)*RR(i+1);
end
C=transpose(C);
```

L13&15_Q73: Compute the series coefficients C_i of Jacobi polynomials for $i=0:10$ and $\alpha = 1$ and $\beta =$

$$2. \quad C_i = \frac{1}{R_{T,j}^{(\alpha,\beta)}} \int_0^T f(x) W_T^{(\alpha,\beta)}(x) P_i^{(\alpha,\beta)}(x) dx$$

% Compute the series coefficients C_i of Jacobi polynomials
% for scale level $M=10$ and $\alpha=1$ and $\beta=2$.

```
function C=JacobiCoeffVector(f,x,M,alpha,beta,T)
```

```
close all;
clear all;
clc;
syms f x M alpha beta T k i j
T=1;
alpha=1;
beta=2;
M=10;
W = (T - x).^alpha * x^beta;
for i=0:M-1
    a1 = (-1)^(i-k);
    a2 = gamma(i + beta+1);
    a3 = gamma(i +k+ alpha + beta +1);
    d1 = gamma(k+beta +1);
    d2 = gamma(i + alpha + beta + 1);
    d3 = gamma(i - k + 1);
    d4 = gamma(k+1);
    d5 = T^k;
    aa = (a1*a2*a3)/(d1*d2*d3*d4*d5);
    P(i+1) = symsum(aa*x^k, k, 0, i);
```

```
end
P;
PW=f*P*W;
for j = 0:M-1
    a1 = T^(alpha + beta +1);
    a2 = gamma(j + alpha + 1);
    a3 = gamma(j + beta + 1);
    d1 = 2*j + alpha + beta + 1;
    d2 = gamma(j + 1);
    d3 = gamma(j + alpha + beta + 1);
    R(j + 1) = (a1*a2*a3)/(d1*d2*d3);
end
R ;
RR=1./R;
for i = 0:M-1
    C(i+1)=int(PW(i+1),x,0,i)*RR(i+1);
end
```




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`C=transpose(C);`

L13&15_Q74: Compute the series coefficients C_i of Jacobi polynomials for scale level $M=6$ and $\alpha = 2$ and $\beta = 1$. While

$$C_i = \frac{1}{R_{T,j}^{(\alpha,\beta)}} \int_0^T f(x) W_T^{(\alpha,\beta)}(x) P_i^{(\alpha,\beta)}(x) dx$$

% Compute the series coefficients C_i of Jacobi polynomials
% for scale level $M=6$ and $\alpha=2$ and $\beta=1$.

`function C=JacobiCoeffVector(f,x,M,alpha,beta,T)`

`close all;`

`clear all;`

`clc;`

`syms f x M alpha beta T k i j`

`T=1;`

`alpha=2;`

`beta=1;`

`M=6;`

`W = (T - x).^alpha * x^beta;`

`for i=0:M-1`

`a1 = (-1)^(i-k);`

`a2 = gamma(i + beta+1);`

`a3 = gamma(i + k+ alpha + beta +1);`

`d1 = gamma(k+beta +1);`

`d2 = gamma(i + alpha + beta + 1);`

`d3 = gamma(i - k + 1);`

`d4 = gamma(k+1);`

`d5 = T^k;`

`aa = (a1*a2*a3)/(d1*d2*d3*d4*d5);`

`P(i+1) = symsum(aa*x^k, k, 0, i);`

`end`

`P;`

`PW=f*P*W;`

`for j = 0:M-1`

`a1 = T^(alpha + beta +1);`

`a2 = gamma(j + alpha + 1);`

`a3 = gamma(j + beta + 1);`

`d1 = 2*j + alpha + beta + 1;`

`d2 = gamma(j + 1);`

`d3 = gamma(j + alpha + beta + 1);`

`R(j + 1) = (a1*a2*a3)/(d1*d2*d3);`

`end`

`R ;`

`RR=1./R;`

`for i = 0:M-1`

`C(i+1)=int(PW(i+1),x,0,i)*RR(i+1);`

`end`

`C=transpose(C);`

L14_Q75: Write MATLAB code to compute the Legendre Polynomials $P_0(z) = 1, P_1(z) = z, P_2(z) = \frac{1}{2}(3z^2 - 1), P_3(z) = \frac{1}{2}(5z^3 - 3z)$ and $P_4(z) = \frac{1}{8}(35z^4 - 30z^2 + 3)$ using the relation $L_{i+1}(Z) = \frac{2i+1}{i+1}zL_i(Z) - \frac{i}{i+1}L_{i-1}(Z).$

% Write MATLAB code to compute the Legendre Polynomials using the relation

`function P=LegenderFunctionVector(M,T,value1,value2)`

`clear all`

`close all`

`clc`

`syms z`

`format short`

`M=2;`

`value(1)=z;`



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```
value(2)=sym(3/2)*z^2-sym(1/2);  
for i=1:M  
    value(i+2)=((2*i+3)/(i+2))*z*value(i+1)-((i+1)/(i+2))*value(i);  
end  
P=simplify(transpose(value));
```

L14_Q76: Write MATLAB code to compute the Legendre Polynomials using the relation $P_0(z) = 1$, $P_1(z) = z$, $P_2(z) = \frac{1}{2}(3z^2 - 1)$, $P_3(z) = \frac{1}{2}(5z^3 - 3z)$ and $P_4(z) = \frac{1}{8}(35z^4 - 30z^2 + 3)$ for $i=1:20$ and using the relation $L_{i+1}(Z) = \frac{2i+1}{i+1}zL_i(Z) - \frac{i}{i+1}L_{i-1}(Z)$.

```
% Write MATLAB code to compute the Legendre Polynomials using the relation  
% for i=1:20  
function P=LegenderFunctionVector(M,T,value1,value2)  
clear all  
close all  
clc  
syms z  
format short  
M=20;  
value(1)=z;  
value(2)=sym(3/2)*z^2-sym(1/2);  
for i=1:M  
    value(i+2)=((2*i+3)/(i+2))*z*value(i+1)-((i+1)/(i+2))*value(i);  
end  
P=simplify(transpose(value));
```

L14_Q77: Write MATLAB code to compute the Legendre Polynomials using the relation $P_0(z) = 1$, $P_1(z) = z$, $P_2(z) = \frac{1}{2}(3z^2 - 1)$, $P_3(z) = \frac{1}{2}(5z^3 - 3z)$ and $P_4(z) = \frac{1}{8}(35z^4 - 30z^2 + 3)$ at $z=0$, $z=1$ and $z=5$ and using the relation $L_{i+1}(Z) = \frac{2i+1}{i+1}zL_i(Z) - \frac{i}{i+1}L_{i-1}(Z)$.

```
% Write MATLAB code to compute the Legendre Polynomials using the relation  
% for i=1:20 and at z=0, z=1 and z=5  
function p=LegenderFunctionVector(M,T,value1,value2)  
clear all;  
close all;  
clc;  
syms z  
z=0;  
%ALSO FIND RESULTS BY PUTTING Z=1 AND Z=5  
format short  
M=20;  
value(1)=z;  
value(2)=sym(3/2)*z^2-sym(1/2);  
for i=1:M  
    value(i+2)=((2*i+3)/(i+2))*z*value(i+1)-((i+1)/(i+2))*value(i);  
end  
p=transpose(value);
```

L14_Q78: Construct the function vector $P_n(x)$ (Orthogonal Shifted Legendre Polynomials) defined on $[0,T]$ of degree 0,1,2 and 3 using the relation $p_i(x) = \sum_{k=0}^i (-1)^{i+k} \frac{\Gamma(i+k+1)}{\Gamma(i-k+1)\Gamma(k+1)^2}$ where $i = 0,1,2,3, \dots$

```
% Construct the function vector Orthogonal Shifted Legendre Polynomials  
% defined on [0,T] of degree 0,1,2 and 3  
function p=FunctionVectorOrthogonalShiftedLegendre(M,T)  
clear all; close all;  
clc;  
syms x T k i;  
M=4;  
for i=0:M-1  
    a1=(-1)^(i+k);  
    a2=gamma(i+k+1);  
    d1=gamma(i-k+1);
```



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```
d2=(gamma(k+1))^2;  
aa=(a1*a2)/(d1*d2);  
p(i+1)=symsum(aa*x^k,k,0,i);
```

```
end  
p=transpose(p);
```

L14_Q79: Construct the function vector $P_n(x)$ (Orthogonal Shifted Legendre Polynomials) defined on $[0,T]$ for $i=0:10$ using the relation $p_i(x) = \sum_{k=0}^i (-1)^{i+k} \frac{\Gamma(i+k+1)}{\Gamma(i-k+1)(\Gamma(k+1))^2}$ where $i = 0,1,2,3, \dots$

% Construct the function vector Orthogonal Shifted Legendre Polynomials
% defined on $[0,T]$ for $i=0:10$

```
function p=FunctionVectorOrthogonalShiftedLegendre(M,T)  
clear all;  
close all;  
clc;  
syms x T k i;  
for i=0:10  
    a1=(-1)^(i+k);  
    a2=gamma(i+k+1);  
    d1=gamma(i-k+1);  
    d2=(gamma(k+1))^2;  
    aa=(a1*a2)/(d1*d2);  
    p(i+1)=symsum(aa*x^k,k,0,i);  
end  
p=transpose(p);
```

L14_Q80: Construct the function vector $P_n(x)$ (Orthogonal Shifted Legendre Polynomials) defined on $[0,T]$ for the scale level $M=15$ using the relation $p_i(x) = \sum_{k=0}^i (-1)^{i+k} \frac{\Gamma(i+k+1)}{\Gamma(i-k+1)(\Gamma(k+1))^2}$ where $i = 0,1,2,3, \dots$

% Construct the function vector Orthogonal Shifted Legendre Polynomials
% defined on $[0,T]$ for $M=15$

```
function p=FunctionVectorOrthogonalShiftedLegendre(M,T)  
clear all;  
close all;  
clc;  
syms x T k i;  
M=15;  
for i=0:M-1  
    a1=(-1)^(i+k);  
    a2=gamma(i+k+1);  
    d1=gamma(i-k+1);  
    d2=(gamma(k+1))^2;  
    aa=(a1*a2)/(d1*d2);  
    p(i+1)=symsum(aa*x^k,k,0,i);  
end  
p=transpose(p);
```

L14&15_Q81: Construct the function vector $P_n(x)$ (Orthogonal Shifted Legendre Polynomials) defined on $[0,1]$ for $i=0:20$ using the relation $p_i(x) = \sum_{k=0}^i (-1)^{i+k} \frac{\Gamma(i+k+1)}{\Gamma(i-k+1)(\Gamma(k+1))^2}$ where $i = 0,1,2,3, \dots$
Moreover compute the values at of $P_n(x)$ at $x=2$.

% Construct the function vector Orthogonal Shifted Legendre Polynomials
% defined on $[0,1]$ for $i=0:20$ using the relation at $x=2$

```
function p=FunctionVectorOrthogonalShiftedLegendre(M,T)  
clear all;  
close all;  
clc;  
syms x T k i;  
M=20;  
T=1;  
x=2;  
for i=0:M  
    a1=(-1)^(i+k);
```




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```
a2=gamma(i+k+1);  
d1=gamma(i-k+1);  
d2=(gamma(k+1))^2;  
aa=(a1*a2)/(d1*d2);  
p(i+1)=symsum(aa*x^k,k,0,i);  
end  
p=transpose(p);
```

L14&15_Q82: Construct the function vector $P_n(x)$ (Orthogonal Shifted Legendre Polynomials) defined on $[0,1]$ for the scale level $M=30$ using the relation $p_i(x) = \sum_{k=0}^i (-1)^{i+k} \frac{\Gamma(i+k+1)}{\Gamma(i-k+1)(\Gamma(k+1))^2}$ where $i = 0,1,2,3,\dots$. Moreover compute the values at of $P_n(x)$ at $x=0$.

% Construct the function vector Orthogonal Shifted Legendre Polynomials
% defined on $[0,1]$ for scale level $M=30$ using the relation at $x=0$;

```
function p=FunctionVectorOrthogonalShiftedLegendre(M,T)  
clear all;  
close all;  
clc;  
syms x T k i;  
M=30;  
T=1;  
x=0;  
for i=0:M-1  
    a1=(-1)^(i+k);  
    a2=gamma(i+k+1);  
    d1=gamma(i-k+1);  
    d2=(gamma(k+1))^2;  
    aa=(a1*a2)/(d1*d2);  
    p(i+1)=symsum(aa*x^k,k,0,i);  
end  
p=transpose(p);
```

L14_Q83: Write MATLAB code for generating the constant factor R_j .

% Write MATLAB code for generating the constant factor
% and using the relation $R_j=1/(2j+1)$

```
function p=ConstantFactor(j,M)  
close all;  
clear all;  
clc;  
M=4;  
format short;  
syms j  
for j=0:M-1  
    a1=1;  
    a2=2*j+1;  
    R(j+1)=a1/a2;  
end  
R
```

L14_Q84: Compute the constant factor defined in R_j for $j=0:6$ and using the relation $R_j = \frac{1}{2j+1}$

% Compute the constant factor defined in R_j for $j=0:6$
% and using the relation $R_j=1/(2j+1)$

```
function p=ConstantFactor(j)  
close all;  
clear all;  
clc;  
format short;  
syms j  
for j=0:6  
    a1=1;  
    a2=2*j+1;  
    R(j+1)=a1/a2;  
end
```



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R

L14&15_Q85: Compute the constant factor defined in R_j for $j=0:10$ and using the relation $R_j = \frac{1}{2j+1}$

% Compute the constant factor defined in R_j for $j=0:10$

% and using the relation $R_j=1/(2j+1)$

```
function p=ConstantFactor(j)
```

```
close all;
```

```
clear all;
```

```
clc;
```

```
format short;
```

```
syms j
```

```
for j=0:10
```

```
    a1=1;
```

```
    a2=2*j+1;
```

```
    R(j+1)=a1/a2;
```

```
end
```

R

L14_Q86: Compute the constant factor defined in R_j for choosing scale level $M=10$ and using the relation $R_j = \frac{1}{2j+1}$

% Write MATLAB code for generating the constant factor

% for choosing scale level $M=10$ and using the relation $R_j=1/(2j+1)$

```
function p=ConstantFactor(j,M)
```

```
close all;
```

```
clear all;
```

```
clc;
```

```
M=10;
```

```
format short;
```

```
syms j
```

```
for j=0:M-1
```

```
    a1=1;
```

```
    a2=2*j+1;
```

```
    R(j+1)=a1/a2;
```

```
end
```

R

L14&15_Q87: Compute the constant factor defined in R_j for choosing scale level $M=40$ and using the relation $R_j = \frac{1}{2j+1}$

% Write MATLAB code for generating the constant factor

% for choosing scale level $M=40$ and using the relation $R_j=1/(2j+1)$

```
function p=ConstantFactor(j,M)
```

```
close all;
```

```
clear all;
```

```
clc;
```

```
M=40;
```

```
format short;
```

```
syms j
```

```
for j=0:M-1
```

```
    a1=1;
```

```
    a2=2*j+1;
```

```
    R(j+1)=a1/a2;
```

```
end
```

R

L14_Q88: Write MATLAB code to compute the series coefficients C_i , $i=0, 1, 2, 3, 4$.

%Write MATLAB code to compute the series coefficients C_i , $i=0, 1, 2, 3, 4$.

```
function P=Lagdrfunvec(M,T)
```

```
clc; clear all; close all;
```

```
syms x T k
```

```
M=4;
```

```
for i=0:M
```

```
    a1=(-1)^(i+k);
```



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```
a2=gamma(i+k+1);
d1=gamma(i-k+1);
d2=(gamma(k+1))^2;
aa=(a1*a2)/(d1*d2);
P(i+1)=symsum(aa*x^k,k,0,i);
end
P=transpose(P);
function R=Lagendrorthogonal(i,M)
syms i
M=4;
for i=0:M-1
a1=1;
a2=2*i+1;
R(i+1)=a1/a2;
end
R
function C=Lagendrcoefvec(i,T,f)
syms x T i f
P=LagdrFunVec(T);
R=LagdrOrthogonal(i);
M=4;
Kw=f*P;
for i=0:M
aa=Kw(i+1);
C(i+1)=(1/(R(i+1)))*int(aa,x,0,T);
end
C=transpose(C);
```

L14_Q89: Compute the series coefficients C_i for $i=0:6$ and using the relation $C_i = \frac{1}{R_j} \int_0^1 f(x) P_i(x) dx$

%Write MATLAB code to compute the series coefficients for $i=0:6$.

```
function P=Lagdrfunvec(M,T)
clc;
clear all;
close all;
syms x T k
M=6;
for i=0:M
a1=(-1)^(i+k);
a2=gamma(i+k+1);
d1=gamma(i-k+1);
d2=(gamma(k+1))^2;
aa=(a1*a2)/(d1*d2);
P(i+1)=symsum(aa*x^k,k,0,i);
end
P=transpose(P);
function R=Lagendrorthogonal(i,M)
syms i
M=6;
for i=0:M-1
a1=1;
a2=2*i+1;
R(i+1)=a1/a2;
end
R
function C=Lagendrcoefvec(i,T,f)
syms x T i f
P=LagdrFunVec(T);
R=LagdrOrthogonal(i);
M=6;
Kw=f*P;
for i=0:M
aa=Kw(i+1);
```




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```
C(i+1)=(1/(R(i+1)))*int(aa,x,0,T);
```

```
end
```

```
C=transpose(C);
```

L14_Q90: Compute the series coefficients C_i for choosing scale level $M=10$ and using the relation

$$C_i = \frac{1}{R_j} \int_0^1 f(x) P_i(x) dx$$

%Write MATLAB code to compute the series coefficients for scale level $M=10$.

```
function P=Lagdrfunvec(M,T)
```

```
clc;
```

```
clear all;
```

```
close all;
```

```
syms x T k
```

```
M=10;
```

```
for i=0:M-1
```

```
    a1=(-1)^(i+k);
```

```
    a2=gamma(i+k+1);
```

```
    d1=gamma(i-k+1);
```

```
    d2=(gamma(k+1))^2;
```

```
    aa=(a1*a2)/(d1*d2);
```

```
    P(i+1)=symsum(aa*x^k,k,0,i);
```

```
end
```

```
P=transpose(P);
```

```
function R=Lagendrorthogonal(i,M)
```

```
syms i
```

```
M=10;
```

```
for i=0:M-1
```

```
    a1=1;
```

```
    a2=2*i+1;
```

```
    R(i+1)=a1/a2;
```

```
end
```

```
R
```

```
function C=Lagendrcoefvec(i,T,f)
```

```
syms x T i f
```

```
P=LagdrFunVec(T);
```

```
R=LagdrOrthogonal(i);
```

```
M=10;
```

```
Kw=f*P;
```

```
for i=0:M
```

```
    aa=Kw(i+1);
```

```
    C(i+1)=(1/(R(i+1)))*int(aa,x,0,T);
```

```
end
```

```
C=transpose(C);
```

L14_Q91: Determined the Legendre Polynomials using the relation $L_{i+1}(Z) = \frac{2i+1}{i+1} zL_i(Z) - \frac{i}{i+1} L_{i-1}(Z)$

For $i=1:10$. That is we have to determined $P_3(z), P_4(z), P_5(z) \dots, P_{10}(z)$ using, $P_1(z) = z$ and $P_2(z) =$

$$\frac{3z^2}{2} - \frac{1}{2}$$

% Write MATLAB code to compute the Legendre Polynomials using the relation

```
% For i=1:10
```

```
function p=LegenderFunctionVector(M,T,value1,value2)
```

```
clear all;
```

```
close all;
```

```
clc;
```

```
syms z
```

```
format short
```

```
M=10;
```

```
value(1)=z;
```

```
value(2)=sym(3/2)*z^2-sym(1/2);
```

```
for i=1:M
```

```
    value(i+2)=((2*i+3)/(i+2))*z*value(i+1)-((i+1)/(i+2))*value(i);
```

```
end
```

```
p=transpose(value);
```



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SOLVED QUESTIONS FOR FINAL TERM EXAM [LECT 09 TO 15]

L14_Q92: Compute the values of the Legendre Polynomials using the relation $L_{i+1}(Z) = \frac{2i+1}{i+1} zL_i(Z) - \frac{i}{i+1} L_{i-1}(Z)$ at $z = 3, z = 4$ and $z = 6$

% Write MATLAB code to compute the Legendre Polynomials using the relation
% at $z=3, z=4$ and $z=6$

```
function p=LegenderFunctionVector(M,T,value1,value2)
clear all;
close all;
clc;
syms z
z=3;
%ALSO FIND RESULTS BY PUTTING Z=4 AND Z=6
format short
M=10;
value(1)=z;
value(2)=sym(3/2)*z^2-sym(1/2);
for i=1:M
    value(i+2)=((2*i+3)/(i+2))*z*value(i+1)-((i+1)/(i+2))*value(i);
end
p=transpose(value);
```

L15_Q93: Construct the one-dimensional Orthogonal Shifted Legendre function vector $P_i(x)$ defined on $[0,1]$ for $i=0:20$ using the relation $p_i(x) = \sum_{k=0}^i (-1)^{i+k} \frac{\Gamma(i+k+1)}{\Gamma(i-k+1)\Gamma(k+1)^2} x^k$ where $i = 0,1,2,3, \dots$

% Construct the function vector Orthogonal Shifted Legendre Polynomials
% defined on $[0,1]$ for $i=0:20$ using the relation;

```
function p=FunctionVectorOrthogonalShiftedLegendre(M,T)
clear all;
close all;
clc;
syms x T k i;
T=1;
M=20;
for i=0:M
    a1=(-1)^(i+k);
    a2=gamma(i+k+1);
    d1=gamma(i-k+1);
    d2=(gamma(k+1))^2;
    aa=(a1*a2)/(d1*d2);
    p(i+1)=symsum(aa*x^k,k,0,i);
end
p=transpose(p);
```

L15_Q94: Construct the one-dimensional Orthogonal Shifted Jacobi function vector $p_n^{(\alpha,\beta)}(x)$ defined on $[0,1]$ for $i=0:19$ and by choosing the values of parameters $\alpha = 0$ and $\beta = 0$ and using equation

$$p_n^{(\alpha,\beta)}(x) = \sum_{k=0}^n \frac{(-1)^{(i-k)} \Gamma(i+\beta+1) \Gamma(i+k+\alpha+\beta+1)}{\Gamma(k+\beta+1) \Gamma(i+\alpha+\beta+1) \Gamma(i-k+1) \Gamma(k+1) T^k} x^k \text{ where } i = 0,1,2,3, \dots$$

% Construct the one-dimensional Orthogonal Shifted Jacobi function vector
% defined on $[0,1]$ for $i=0:19$ and by choosing the values of parameters $\alpha=0$ and $\beta=0$

```
function P=JacobFunVec(alpha,beta,M)
clear all
close all
clc
syms x T k
M=19;
T=1;
alpha=0;
beta=0;
for i=0:M-1
    a1=(-1)^(i-k);
    a2=gamma(i+beta+1);
```



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SOLVED QUESTIONS FOR FINAL TERM EXAM [LECT 09 TO 15]



```
a3=gamma(i+k+alpha+beta+1);  
d1=gamma(k+beta+1);  
d2=gamma(i+alpha+beta+1);  
d3=gamma(i-k+1);  
d4=gamma(k+1);  
d5=T^k;  
aa=(a1*a2*a3)/(d1*d2*d3*d4*d5);  
P(i+1)=symsum(aa*x^k,k,0,i);  
end  
P=transpose(P);
```

L15_Q95: Compute the following relation for $j=0:20$ and choosing the values of $a=1$ and $b=2$:

$$\Omega_j^{(a,b)}(x) = \frac{(4)^{a+b+1} \Gamma(j+3a+4b+1)(j+9)!}{(2j+a+b+1) \Gamma(j+a+5)(j+10a+b+1)!}$$

% Compute the following relation for $j=0:20$ and choosing the values of $a=1$ and $b=2$:

```
function P=JacobFunVec(a,b,j)  
clear all;  
close all;  
clc;  
syms x k  
i=20;  
a=1;  
b=2;  
for j=0:i  
    a1=(4)^(a+b+1);  
    a2=gamma(j+3*a+4*b+1);  
    a3=factorial(j+9);  
    d1=2*j+a+b+1;  
    d2=gamma(j+a+5);  
    d3=factorial(j+10*a+b+1);  
    aa=(a1*a2*a3)/(d1*d2*d3);  
    P(j+1)=aa;  
end  
P=transpose(P);
```

DON'T LOOK FOR SOMEONE WHO CAN SOLVE YOUR PROBLEMS,

INSTEAD GO AND STAND IN FRONT OF THE MIRROR,

LOOK STRAIGHT INTO YOUR EYES,

AND YOU WILL SEE THE BEST PERSON WHO CAN SOLVE YOUR PROBLEMS!

ALWAYS TRUST YOURSELF.

The End